

## Mark Scheme

Q1.

| Part | Working or answer an examiner might expect to see                                                                                                         | Mark | Notes                                                                                                |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------|------------------------------------------------------------------------------------------------------|
| (a)  | $V = Ae^{-kt}$                                                                                                                                            | M1   | This mark is given for suggesting a suitable exponential model for $V$ in terms of $t$               |
|      | When $t = 0$ and $V = 20\,000$ , $A = 20\,000$                                                                                                            | M1   | This mark is given for using the model to show the initial value for $A$ is £20 000                  |
|      | When $t = 1$ and $V = 16\,000$ ,<br>$16\,000 = 20\,000e^{-1k}$<br>$k = \ln 0.8 = -0.223$                                                                  | M1   | This mark is given for using the value of the car after one year to find a value for $k$             |
|      | $V = 20\,000e^{-0.223t}$                                                                                                                                  | A1   | This mark is given for finding a fully correct exponential model                                     |
| (b)  | When $t = 10$ , $V = £2150$                                                                                                                               | M1   | This mark is given for finding a value for $V$ when $t = 10$                                         |
|      | This model is reliable since the value £2150 is close to £2000                                                                                            | A1   | This mark is given for a valid statement comparing the two possible values of the car after 10 years |
| (c)  | For example:<br>The value of $k$ should be increased (e.g. $V = 20\,000e^{-0.1t}$ )<br>A constant should be added (e.g. $V = 20\,000e^{-0.223t} + 2000$ ) | B1   | This mark is given for a statement suggesting a valid adaptation                                     |

Q2.

| Part | Working or answer an examiner might expect to see                                                                                           | Mark | Notes                                                                                     |
|------|---------------------------------------------------------------------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------------|
| (a)  | $\log a - \log b = \log \frac{a}{b}$                                                                                                        | B1   | This mark is given for restating the log equation using $\log \frac{a}{b}$                |
|      | $a - b = \frac{a}{b}$<br>$ab - b^2 = a$<br>$ab - a = b^2$                                                                                   | M1   | This mark is given for rearranging so that terms in a are on one side of the equation     |
|      | $a(b - 1) = b^2$<br>$a = \frac{b^2}{(b - 1)}$                                                                                               | A1   | This mark is for rearranging to show the result required                                  |
| (b)  | $b \neq 1$                                                                                                                                  | B1   | This mark is given for deducing that $b \neq 1$                                           |
|      | <p>Since <math>a &gt; 0</math>, <math>\frac{b^2}{(b - 1)} &gt; 0</math></p> <p><math>b &gt; 1</math> since <math>b^2</math> is positive</p> | B1   | This mark is given for stating that $b > 1$ and explaining the reason for the restriction |

Q3.

| Question                                                                                                                                                                                                                                                                                                                                                                                                                                               | Scheme                                                                                  | Marks | AOs  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-------|------|
| (a)                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $gf(x) = 3 \ln e^x$                                                                     | M1    | 1.1b |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $= 3x, (x \in \mathbb{R})$                                                              | A1    | 1.1b |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                         | (2)   |      |
| (b)                                                                                                                                                                                                                                                                                                                                                                                                                                                    | $gf(x) = fg(x) \Rightarrow 3x = x^3$                                                    | M1    | 1.1b |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $\Rightarrow x^3 - 3x = 0 \Rightarrow x =$                                              | M1    | 1.1b |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                        | $\Rightarrow x = (+)\sqrt{3}$ only as $\ln x$ is not defined at $x = 0$ and $-\sqrt{3}$ | M1    | 2.2a |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                         | (3)   |      |
| (5 marks)                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                         |       |      |
| <b>Notes:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                         |       |      |
| (a)                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                         |       |      |
| M1: For applying the functions in the correct order                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                         |       |      |
| A1: The simplest form is required so it must be $3x$ and not left in the form $3 \ln e^x$<br>An answer of $3x$ with no working would score both marks                                                                                                                                                                                                                                                                                                  |                                                                                         |       |      |
| (b)                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                         |       |      |
| M1: Allow the candidates to score this mark if they have $e^{3 \ln x} = \text{their } 3x$                                                                                                                                                                                                                                                                                                                                                              |                                                                                         |       |      |
| M1: For solving their cubic in $x$ and obtaining at least one solution.                                                                                                                                                                                                                                                                                                                                                                                |                                                                                         |       |      |
| A1: For either stating that $x = \sqrt{3}$ only as $\ln x$ (or $3 \ln x$ ) is not defined at $x = 0$ and $-\sqrt{3}$<br>or stating that $3x = x^3$ would have three answers, one positive one negative and one zero but $\ln x$ (or $3 \ln x$ ) is not defined for $x \leq 0$ so therefore there is only one (real) answer.<br>Note: Student who mix up $fg$ and $gf$ can score full marks in part (b) as they have already been penalised in part (a) |                                                                                         |       |      |

Q4.

| Question     | Scheme                                                                                                                                                                                                                                                               | Marks | AOs  |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------|
|              | $N = \frac{900}{3 + 7e^{-0.25t}} = 900(3 + 7e^{-0.25t})^{-1}, t \in \mathbb{R}, t \geq 0; \frac{dN}{dt} = \frac{N(300 - N)}{1200}$                                                                                                                                   |       |      |
| (a)          | 90                                                                                                                                                                                                                                                                   | B1    | 3.4  |
|              |                                                                                                                                                                                                                                                                      | (1)   |      |
| (b)<br>Way 1 | $\frac{dN}{dt} = -900(3 + 7e^{-0.25t})^{-2} (7(-0.25)e^{-0.25t}) \left\{ = \frac{900(0.25)(7)e^{-0.25t}}{(3 + 7e^{-0.25t})^2} \right\}$                                                                                                                              | M1    | 2.1  |
|              |                                                                                                                                                                                                                                                                      | A1    | 1.1b |
|              | $\Rightarrow \frac{dN}{dt} = \frac{900(0.25) \left( \left( \frac{900}{N} - 3 \right) \right)}{\left( \frac{900}{N} \right)^2}$                                                                                                                                       | dM1   | 2.1  |
|              | correct algebra leading to $\frac{dN}{dt} = \frac{N(300 - N)}{1200} *$                                                                                                                                                                                               | A1*   | 1.1b |
|              |                                                                                                                                                                                                                                                                      | (4)   |      |
| (b)<br>Way 2 | $\frac{dN}{dt} = -900(3 + 7e^{-0.25t})^{-2} (7(-0.25)e^{-0.25t}) \left\{ = \frac{900(0.25)(7)e^{-0.25t}}{(3 + 7e^{-0.25t})^2} \right\}$                                                                                                                              | M1    | 2.1  |
|              |                                                                                                                                                                                                                                                                      | A1    | 1.1b |
|              | $\frac{N(300 - N)}{1200} = \frac{\left( \frac{900}{3 + 7e^{-0.25t}} \right) \left( 300 - \frac{900}{3 + 7e^{-0.25t}} \right)}{1200}$                                                                                                                                 | dM1   | 2.1  |
|              | LHS = $\frac{1575e^{-0.25t}}{(3 + 7e^{-0.25t})^2}$ o.e.,<br>RHS = $\frac{900(300(3 + 7e^{-0.25t}) - 900)}{1200(3 + 7e^{-0.25t})^2} = \frac{1575e^{-0.25t}}{(3 + 7e^{-0.25t})^2}$ o.e.<br>and states hence $\frac{dN}{dt} = \frac{N(300 - N)}{1200}$ (or LHS = RHS) * | A1*   | 1.1b |
|              |                                                                                                                                                                                                                                                                      | (4)   |      |
| (c)          | Deduces $N = 150$ (can be implied)                                                                                                                                                                                                                                   | B1    | 2.2a |
|              | so $150 = \frac{900}{3 + 7e^{-0.25t}} \Rightarrow e^{-0.25t} = \frac{3}{7}$                                                                                                                                                                                          | M1    | 3.4  |
|              | $T = -4 \ln\left(\frac{3}{7}\right)$ or $T = \text{awrt } 3.4$ (months)                                                                                                                                                                                              | dM1   | 1.1b |
|              |                                                                                                                                                                                                                                                                      | A1    | 1.1b |
|              |                                                                                                                                                                                                                                                                      | (4)   |      |
| (d)          | either one of 299 or 300                                                                                                                                                                                                                                             | B1    | 3.4  |
|              |                                                                                                                                                                                                                                                                      | (1)   |      |
| (10 marks)   |                                                                                                                                                                                                                                                                      |       |      |

| Notes for Question |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (b)                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| M1:                | Attempts to differentiate using <ul style="list-style-type: none"> <li>the chain rule to give <math>\frac{dN}{dt} = \pm Ae^{-0.25t}(3+7e^{-0.25t})^{-2}</math> or <math>\frac{\pm Ae^{-0.25t}}{(3+7e^{-0.25t})^2}</math> o.e.</li> <li>the quotient rule to give <math>\frac{dN}{dt} = \frac{(3+7e^{-0.25t})(0) \pm Ae^{-0.25t}}{(3+7e^{-0.25t})^2}</math></li> <li>implicit differentiation to give <math>N(3+7e^{-0.25t}) = 900 \Rightarrow (3+7e^{-0.25t})\frac{dN}{dt} \pm ANe^{-0.25t} = 0</math>, o.e. where <math>A \neq 0</math></li> </ul> |
| Note:              | Condone a slip in copying $(3+7e^{-0.25t})$ for the M mark                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| A1:                | A correct differentiation statement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Note:              | Implicit differentiation gives $(3+7e^{-0.25t})\frac{dN}{dt} - 1.75Ne^{-0.25t} = 0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| dM1:               | Way 1: Complete attempt, by eliminating $t$ , to form an equation linking $\frac{dN}{dt}$ and $N$ only<br>Way 2: Complete substitution of $N = \frac{900}{3+7e^{-0.25t}}$ into $\frac{dN}{dt} = \frac{N(300-N)}{1200}$                                                                                                                                                                                                                                                                                                                              |
| Note:              | Way 1: e.g. substitutes $3+7e^{-0.25t} = \frac{900}{N}$ and $e^{-0.25t} = \frac{900}{N} - 3$ or substitutes $e^{-0.25t} = \frac{N}{7} - 3$ into their $\frac{dN}{dt} = \dots$ to form an equation linking $\frac{dN}{dt}$ and $N$                                                                                                                                                                                                                                                                                                                   |
| A1*:               | Way 1: Correct algebra leading to $\frac{dN}{dt} = \frac{N(300-N)}{1200}$ *<br>Way 2: See scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| (c)                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| B1:                | Deduces or shows that $\frac{dN}{dt}$ is maximised when $N = 150$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| M1:                | Uses the model $N = \frac{900}{3+7e^{-0.25t}}$ with their $N = 150$ and proceeds as far as $e^{-0.25T} = k, k > 0$<br>or $e^{0.25T} = k, k > 0$ . Condone $t \equiv T$                                                                                                                                                                                                                                                                                                                                                                              |
| dM1:               | Correct method of using logarithms to find a value for $T$ . Condone $t \equiv T$                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| A1:                | see scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Note:              | $\frac{d^2N}{dt^2} = \frac{dN}{dt} \left( \frac{300}{1200} - \frac{2N}{1200} \right) = 0 \Rightarrow N = 150$ is acceptable for B1                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Note:              | Ignore units for $T$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Note:              | Applying $300 = \frac{900}{3+7e^{-0.25t}} \Rightarrow t = \dots$ or $0 = \frac{900}{3+7e^{-0.25t}} \Rightarrow t = \dots$ is M0 dM0 A0                                                                                                                                                                                                                                                                                                                                                                                                              |
| Note:              | M1 dM1 can only be gained in (c) by using an $N$ value in the range $90 < N < 300$                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| (d)                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| B1:                | 300 (or accept 299)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| Question     | Scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Marks | AOs  |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------|
|              | $N = \frac{900}{3 + 7e^{-0.25t}} = 900(3 + 7e^{-0.25t})^{-1}, t \in \mathbb{R}, t \geq 0; \frac{dN}{dt} = \frac{N(300 - N)}{1200}$                                                                                                                                                                                                                                                                                                                                                         |       |      |
| (b)<br>Way 3 | $\int \frac{1}{N(300 - N)} dN = \int \frac{1}{1200} dt$                                                                                                                                                                                                                                                                                                                                                                                                                                    | M1    | 2.1  |
|              | $\int \frac{1}{300} \left( \frac{1}{N} + \frac{1}{300 - N} \right) dN = \int \frac{1}{1200} dt$                                                                                                                                                                                                                                                                                                                                                                                            | A1    | 1.1b |
|              | $\frac{1}{300} \ln N - \frac{1}{300} \ln(300 - N) = \frac{1}{1200} t \{+c\}$                                                                                                                                                                                                                                                                                                                                                                                                               |       |      |
|              | $\{t = 0, N = 90 \Rightarrow\} c = \frac{1}{300} \ln(90) - \frac{1}{300} \ln(210) \Rightarrow c = \frac{1}{300} \ln\left(\frac{3}{7}\right)$<br>$\frac{1}{300} \ln N - \frac{1}{300} \ln(300 - N) = \frac{1}{1200} t + \frac{1}{300} \ln\left(\frac{3}{7}\right)$<br>$\ln N - \ln(300 - N) = \frac{1}{4} t + \ln\left(\frac{3}{7}\right)$<br>$\ln\left(\frac{N}{300 - N}\right) = \frac{1}{4} t + \ln\left(\frac{3}{7}\right) \Rightarrow \frac{N}{300 - N} = \frac{3}{7} e^{\frac{t}{4}}$ | dM1   | 2.1  |
|              | $7N = 3e^{\frac{t}{4}}(300 - N) \Rightarrow 7N + 3Ne^{\frac{t}{4}} = 900e^{\frac{t}{4}}$<br>$N(7 + 3e^{\frac{t}{4}}) = 900e^{\frac{t}{4}} \Rightarrow N = \frac{900e^{\frac{t}{4}}}{7 + 3e^{\frac{t}{4}}} \Rightarrow N = \frac{900}{3 + 7e^{-0.25t}} *$                                                                                                                                                                                                                                   | A1*   | 1.1b |
|              | (4)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       |      |
| (b)<br>Way 4 | $N(3 + 7e^{-0.25t}) = 900 \Rightarrow e^{-0.25t} = \frac{1}{7} \left( \frac{900}{N} - 3 \right) \Rightarrow e^{-0.25t} = \frac{900 - 3N}{7N}$<br>$\Rightarrow t = -4(\ln(900 - 3N) - \ln(7N))$<br>$\Rightarrow \frac{dt}{dN} = -4 \left( \frac{-3}{900 - 3N} - \frac{7}{7N} \right)$                                                                                                                                                                                                       | M1    | 2.1  |
|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | A1    | 1.1b |
|              | $\frac{dt}{dN} = 4 \left( \frac{1}{300 - N} + \frac{1}{N} \right) \Rightarrow \frac{dt}{dN} = 4 \left( \frac{N + 300 - N}{N(300 - N)} \right)$                                                                                                                                                                                                                                                                                                                                             | dM1   | 2.1  |
|              | $\frac{dt}{dN} = \left( \frac{1200}{N(300 - N)} \right) \Rightarrow \frac{dN}{dt} = \frac{N(300 - N)}{1200} *$                                                                                                                                                                                                                                                                                                                                                                             | A1*   | 1.1b |
|              | (4)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       |      |

| Notes for Question Continued |                                                                                                                                                                                                              |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (b)<br>Way 3                 |                                                                                                                                                                                                              |
| M1:                          | Separates the variables, an attempt to form and apply partial fractions and integrates to give $\ln$ terms = $kt \{+c\}$ , $k \neq 0$ , with or without a constant of integration $c$                        |
| A1:                          | $\frac{1}{300} \ln N - \frac{1}{300} \ln(300 - N) = \frac{1}{1200} t \{+c\}$ or equivalent with or without a constant of integration $c$                                                                     |
| dM1:                         | Uses $t = 0, N = 90$ to find their constant of integration and obtains an expression of the form $\lambda e^{\frac{t}{4}} = f(N)$ ; $\lambda \neq 0$ or $\lambda e^{-\frac{t}{4}} = f(N)$ ; $\lambda \neq 0$ |
| A1*:                         | Correct manipulation leading to $N = \frac{900}{3 + 7e^{-0.25t}} *$                                                                                                                                          |
| (b)<br>Way 4                 |                                                                                                                                                                                                              |
| M1:                          | Valid attempt to make $t$ the subject, followed by an attempt to find two $\ln$ derivatives, condoning sign errors and constant errors.                                                                      |
| A1:                          | $\frac{dt}{dN} = -4 \left( \frac{-3}{900 - 3N} - \frac{7}{7N} \right)$ or equivalent                                                                                                                         |
| dM1:                         | Forms a common denominator to combine their fractions                                                                                                                                                        |
| A1*:                         | Correct algebra leading to $\frac{dN}{dt} = \frac{N(300 - N)}{1200} *$                                                                                                                                       |



Q5.

| Question  | Scheme                                                                                                                                                                      | Marks               | AOs  |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------|
| (a)       | $t = 0, \theta = 18 \Rightarrow 18 = A - B$<br>or<br>$t = 10, \theta = 44 \Rightarrow 44 = A - Be^{-0.7}$                                                                   | M1                  | 3.1b |
|           | $t = 0, \theta = 18 \Rightarrow 18 = A - B$<br>and<br>$t = 10, \theta = 44 \Rightarrow 44 = A - Be^{-0.7}$<br>and<br>$\Rightarrow A = \dots, B = \dots$                     | M1                  | 3.1a |
|           | At least one of: $A = 69.6, B = 51.6$ but allow awrt 70/awrt 52                                                                                                             | A1<br>M1 on<br>EPEN | 1.1b |
|           | $\theta = 69.6 - 51.6e^{-0.07t}$                                                                                                                                            | A1                  | 3.3  |
|           |                                                                                                                                                                             | (4)                 |      |
| (b)       | The maximum temperature is "69.6"(°C) (according to the model)<br>(The model has an) upper limit of "69.6"(°C)<br>(The model suggests that) the boiling point is "69.6"(°C) | B1ft                | 3.4  |
|           | Model is not appropriate as 69.6(°C) is much lower than 78(°C)                                                                                                              | B1ft                | 3.5a |
|           |                                                                                                                                                                             | (2)                 |      |
| (6 marks) |                                                                                                                                                                             |                     |      |

**Notes:**

- (a)  
**M1:** Makes the first key step in the solution of the problem. Substitutes  $t = 0$  and  $\theta = 18$  or  $t = 10$  and  $\theta = 44$  into the equation of the model to obtain an equation connecting  $A$  and  $B$ .  
 Note that  $18 = A - Be^0$  scores M0 unless  $18 = A - B$  is seen or implied later.  
 If they do not obtain an equation in  $A$  and  $B$  using the first conditions e.g. they have  $18 = A - 1$  then they can score this mark if they substitute  $A = 19$  directly into  $44 = A - Be^{-0.7}$  as an equation in  $A$  and  $B$  is implied.  
**M1:** Substitutes  $t = 0$  and  $\theta = 18$  and  $t = 10$  and  $\theta = 44$  to obtain 2 equations connecting  $A$  and  $B$  and then proceeds to solve their equations in  $A$  and  $B$  simultaneously to obtain values for both constants. Do not be too concerned with the processing as long as values for  $A$  and  $B$  are obtained.  
**A1(M1 on EPEN):** For  $A =$  awrt 70 or  $B =$  awrt 52  
**A1:** For  $\theta = 69.6 - 51.6e^{-0.07t}$  **Must be a fully correct equation as shown but allow recovery if seen in (b).**  
 Note that some candidates evaluate  $e^0$  as 0 and so obtain  $A = 18$  and then write  $44 = 18 - Be^{-0.7}$  and solve for  $B$ . Such attempts can score M1M0A0A0 only.  
 (b)  
**B1ft:** Identifies  $A$  as the boiling point/maximum temperature in the model. Follow through their  $A$ .  
**B1ft:** Makes a valid conclusion (valid/not valid, good/not good etc.) that refers to the 78 and includes a reference to a significant/large difference

Alternative provided their  $A < 78$

- B1ft:**  $\theta = 69.6 - 51.6e^{-0.07t} = 78 \Rightarrow 51.6e^{-0.07t} = 69.6 - 78 = -8.4$   
 $\Rightarrow e^{-0.07t} = -\frac{7}{43}$  and  $\ln\left(-\frac{7}{43}\right)$  and makes a reference to the fact that the equation cannot be solved or e.g. cannot take log of a negative number. You can condone numerical slips in the calculation.  
**B1ft:** Model is not appropriate as 69.6(°C) is much lower than 78(°C)  
**Minimum for both marks:** The model is not appropriate as "69.6"(°C) is much lower than 78(°C)  
 Note that these marks are not available if their equation is solvable. Note also that B0B1 is not possible.

Q6.

| Question   | Scheme                                                                                                                                                                                                                                            | Marks | AOs  |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------|
| (a)        | Sets $\frac{1}{P(11-2P)} = \frac{A}{P} + \frac{B}{(11-2P)}$                                                                                                                                                                                       | B1    | 1.1a |
|            | Substitutes either $P=0$ or $P=\frac{11}{2}$ into<br>$1 = A(11-2P) + BP \Rightarrow A \text{ or } B$                                                                                                                                              | M1    | 1.1b |
|            | $\frac{1}{P(11-2P)} = \frac{1/11}{P} + \frac{2/11}{(11-2P)}$                                                                                                                                                                                      | A1    | 1.1b |
|            |                                                                                                                                                                                                                                                   | (3)   |      |
| (b)        | Separates the variables<br>$\int \frac{22}{P(11-2P)} dP = \int 1 dt$                                                                                                                                                                              | B1    | 3.1a |
|            | Uses (a) and attempts to integrate $\int \frac{2}{P} + \frac{4}{(11-2P)} dP = t + c$                                                                                                                                                              | M1    | 1.1b |
|            | $2 \ln P - 2 \ln(11-2P) = t + c$                                                                                                                                                                                                                  | A1    | 1.1b |
|            | Substitutes $t=0, P=1 \Rightarrow t=0, P=1 \Rightarrow c = (-2 \ln 9)$                                                                                                                                                                            | M1    | 3.1a |
|            | Substitutes $P=2 \Rightarrow t = 2 \ln 2 + 2 \ln 9 - 2 \ln 7$                                                                                                                                                                                     | M1    | 3.1a |
|            | Time = 1.89 years                                                                                                                                                                                                                                 | A1    | 3.2a |
|            |                                                                                                                                                                                                                                                   | (6)   |      |
| (c)        | Uses ln laws $2 \ln P - 2 \ln(11-2P) = t - 2 \ln 9$<br>$\Rightarrow \ln\left(\frac{9P}{11-2P}\right) = \frac{1}{2}t$                                                                                                                              | M1    | 2.1  |
|            | Makes 'P' the subject $\Rightarrow \left(\frac{9P}{11-2P}\right) = e^{\frac{1}{2}t}$<br>$\Rightarrow 9P = (11-2P)e^{\frac{1}{2}t}$<br>$\Rightarrow P = f\left(e^{\frac{1}{2}t}\right) \text{ or } \Rightarrow P = f\left(e^{\frac{1}{2}t}\right)$ | M1    | 2.1  |
|            | $\Rightarrow P = \frac{11}{2+9e^{\frac{1}{2}t}} \Rightarrow A=11, B=2, C=9$                                                                                                                                                                       | A1    | 1.1b |
|            |                                                                                                                                                                                                                                                   | (3)   |      |
| (12 marks) |                                                                                                                                                                                                                                                   |       |      |



**Notes:**

(a)

**B1:** Sets  $\frac{1}{P(11-2P)} = \frac{A}{P} + \frac{B}{(11-2P)}$

**M1:** Substitutes  $P=0$  or  $P=\frac{11}{2}$  into  $1=A(11-2P)+BP \Rightarrow A$  or  $B$

Alternatively compares terms to set up and solve two simultaneous equations in  $A$  and  $B$

**A1:**  $\frac{1}{P(11-2P)} = \frac{1/11}{P} + \frac{2/11}{(11-2P)}$  or equivalent  $\frac{1}{P(11-2P)} = \frac{1}{11P} + \frac{2}{11(11-2P)}$

Note: The correct answer with no working scores all three marks.

(b)

**B1:** Separates the variables to reach  $\int \frac{22}{P(11-2P)} dP = \int 1 dt$  or equivalent

**M1:** Uses part (a) and  $\int \frac{A}{P} + \frac{B}{(11-2P)} dP = A \ln P \pm C \ln(11-2P)$

**A1:** Integrates both sides to form a correct equation including a 'c' Eg  $2 \ln P - 2 \ln(11-2P) = t + c$

**M1:** Substitutes  $t=0$  and  $P=1$  to find  $c$

**M1:** Substitutes  $P=2$  to find  $t$ . This is dependent upon having scored both previous M's

**A1:** Time = 1.89 years

(c)

**M1:** Uses correct log laws to move from  $2 \ln P - 2 \ln(11-2P) = t + c$  to  $\ln\left(\frac{P}{11-2P}\right) = \frac{1}{2}t + d$  for their numerical 'c'

**M1:** Uses a correct method to get  $P$  in terms of  $e^{\frac{1}{2}t}$   
 This can be achieved from  $\ln\left(\frac{P}{11-2P}\right) = \frac{1}{2}t + d \Rightarrow \frac{P}{11-2P} = e^{\frac{1}{2}t+d}$  followed by cross multiplication and collection of terms in  $P$  (See scheme)

Alternatively uses a correct method to get  $P$  in terms of  $e^{\frac{1}{2}t}$  For example  
 $\frac{P}{11-2P} = e^{\frac{1}{2}t+d} \Rightarrow \frac{11-2P}{P} = e^{-(\frac{1}{2}t+d)} \Rightarrow \frac{11}{P} - 2 = e^{-(\frac{1}{2}t+d)} \Rightarrow \frac{11}{P} = 2 + e^{-(\frac{1}{2}t+d)}$  followed by division

**A1:** Achieves the correct answer in the form required.  $P = \frac{11}{2 + 9e^{-\frac{1}{2}t}} \Rightarrow A=11, B=2, C=9$  oe

Q7.

| Question                                                                                                                                 | Scheme                                                                               | Marks | AOs  |
|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-------|------|
| (a)                                                                                                                                      | Substitutes $t = 0.5$ into $m = 25e^{-0.05t} \Rightarrow m = 25e^{-0.05 \times 0.5}$ | M1    | 3.4  |
|                                                                                                                                          | $\Rightarrow m = 24.4\text{g}$                                                       | A1    | 1.1b |
|                                                                                                                                          |                                                                                      | (2)   |      |
| (b)                                                                                                                                      | States or uses $\frac{d}{dt}(e^{-0.05t}) = \pm C e^{-0.05t}$                         | M1    | 2.1  |
|                                                                                                                                          | $\frac{dm}{dt} = -0.05 \times 25e^{-0.05t} = -0.05m \Rightarrow k = -0.05$           | A1    | 1.1b |
|                                                                                                                                          |                                                                                      | (2)   |      |
| (4 marks)                                                                                                                                |                                                                                      |       |      |
| <b>Notes:</b>                                                                                                                            |                                                                                      |       |      |
| (a)                                                                                                                                      |                                                                                      |       |      |
| M1: Substitutes $t = 0.5$ into $m = 25e^{-0.05t} \Rightarrow m = 25e^{-0.05 \times 0.5}$                                                 |                                                                                      |       |      |
| A1: $m = 24.4\text{g}$ An answer of $m = 24.4\text{g}$ with no working would score both marks                                            |                                                                                      |       |      |
| (b)                                                                                                                                      |                                                                                      |       |      |
| M1: Applies the rule $\frac{d}{dt}(e^{kt}) = k e^{kt}$ in this context by stating or using $\frac{d}{dt}(e^{-0.05t}) = \pm C e^{-0.05t}$ |                                                                                      |       |      |
| A1: $\frac{dm}{dt} = -0.05 \times 25e^{-0.05t} = -0.05m \Rightarrow k = -0.05$                                                           |                                                                                      |       |      |

Q8.

| Question  | Scheme                                                                                                                                                  | Marks | AOs  |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------|
| (a)       | $T = al^b \Rightarrow \log_{10} T = \log_{10} a + \log_{10} l^b$                                                                                        | M1    | 2.1  |
|           | $\Rightarrow \log_{10} T = \log_{10} a + b \log_{10} l^*$<br>or<br>$\Rightarrow \log_{10} T = b \log_{10} l + \log_{10} a^*$                            | A1*   | 1.1b |
|           |                                                                                                                                                         | (2)   |      |
| (b)       | $b = 0.495$ or $b = \frac{45}{91}$                                                                                                                      | B1    | 2.2a |
|           | $0 = "0.495" \times -0.7 + \log_{10} a \Rightarrow a = 10^{0.346...}$<br>or<br>$0.45 = "0.495" \times 0.21 + \log_{10} a \Rightarrow a = 10^{0.346...}$ | M1    | 3.1a |
|           | $T = 2.22l^{0.495}$                                                                                                                                     | A1    | 3.3  |
|           |                                                                                                                                                         | (3)   |      |
| (c)       | The time taken for one swing of a pendulum of length 1 m                                                                                                | B1    | 3.2a |
|           |                                                                                                                                                         | (1)   |      |
| (6 marks) |                                                                                                                                                         |       |      |

### Notes

(a)

M1: Takes logs of both sides and shows the addition law.

Implied by  $T = al^b \Rightarrow \log_{10} a + \log_{10} l^b$

A1\*: Uses the power law to obtain the given equation with no errors. Allow the bases to be missing in the working but they must be present in the final answer.

Also allow  $t$  rather than  $T$  and  $A$  rather than  $a$ .

**Allow working backwards e.g.**

$$\log_{10} T = b \log_{10} l + \log_{10} a \Rightarrow \log_{10} T = \log_{10} l^b + \log_{10} a$$

$$\Rightarrow \log_{10} T = \log_{10} al^b \Rightarrow T = al^b *$$

M1: Uses the given answer and uses the power law and addition law correctly

A1: Reaches the given equation with no errors as above

(b)

B1: Deduces the correct value for  $b$  (Allow awrt 0.495 or  $\frac{45}{91}$ )

M1: Correct strategy to find the value of  $a$ .

E.g. substitutes one of the given points and their value for  $b$  into  $\log_{10} T = \log_{10} a + b \log_{10} l$  and uses correct log work to identify the value of  $a$ . Allow slips in rearranging their equation but must be correct log work to find  $a$ .

Alternatively finds the equation of the straight line and equates the constant to  $\log_{10} a$  and uses correct log work to identify the value of  $a$ .

E.g.  $y - 0.45 = "0.495"(x - 0.21) \Rightarrow y = "0.495"x + 0.346 \Rightarrow a = 10^{0.346} = \dots$

A1: Complete equation  $T = 2.22l^{0.495}$  or  $T = 2.22l^{\frac{45}{91}}$

(Allow awrt 2.22 and awrt 0.495 or  $\frac{45}{91}$ )

**Must see the equation not just correct values as it is a requirement of the question.**

(c)

B1: Correct interpretation

Q9.

| Part                   | Working or answer an examiner might expect to see                                                                                                       | Mark | Notes                                                                                                       |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------------------------------|
| (i)                    | $\sum_{r=4}^{\infty} 20 \times \left(\frac{1}{2}\right)^r$ $= \sum_{r=1}^{\infty} 20 \times \left(\frac{1}{2}\right)^r - (10 + 5 + 2.5)$                | M1   | This mark is given for a method to find the sum to infinity of a GP                                         |
|                        | $= \frac{10}{1 - \frac{1}{2}} - (10 + 5 + 2.5)$                                                                                                         | M1   | This mark is given for a method to use a correct sum formula with a correct first term                      |
|                        | $= 2.5$                                                                                                                                                 | A1   | This mark is given for a correct value for the sum                                                          |
| (ii)                   | $\sum_{n=1}^{48} \log_5 \left(\frac{n+2}{n+1}\right)$ $= \log_5 \frac{50}{49} + \log_5 \frac{49}{48} + \dots + \log_5 \frac{4}{3} + \log_5 \frac{3}{2}$ | M1   | This mark is given for writing out at least four terms of the sum, including the first two and the last two |
|                        | $= \log_5 \frac{3 \times 4 \times \dots \times 48 \times 49 \times 50}{2 \times 3 \times 4 \times \dots \times 48 \times 49} = \log_5 \frac{50}{2}$     | M1   | This mark is given for using the rules of logs and cancelling terms                                         |
|                        | $= 2$                                                                                                                                                   | A1   | This mark is given for a full proof to show the expression is equal to 2 as required                        |
| <b>(Total 6 marks)</b> |                                                                                                                                                         |      |                                                                                                             |

Q10.

| Question  | Scheme                                                                                                    | Marks    | AOs          |
|-----------|-----------------------------------------------------------------------------------------------------------|----------|--------------|
| (a)       | $\frac{dH}{dt} = \frac{H \cos 0.25t}{40} \Rightarrow \int \frac{1}{H} dH = \int \frac{\cos 0.25t}{40} dt$ | M1       | 3.1a         |
|           | $\ln H = \frac{1}{10} \sin 0.25t (+c)$                                                                    | M1<br>A1 | 1.1b<br>1.1b |
|           | Substitutes $t = 0, H = 5 \Rightarrow c = \ln(5)$                                                         | dM1      | 3.4          |
|           | $\ln\left(\frac{H}{5}\right) = \frac{1}{10} \sin 0.25t \Rightarrow H = 5e^{0.1 \sin 0.25t} *$             | A1*      | 2.1          |
|           |                                                                                                           | (5)      |              |
| (b)       | Max height = $5e^{0.1} = 5.53 \text{ m}$ (Condone lack of units)                                          | B1       | 3.4          |
|           |                                                                                                           | (1)      |              |
| (c)       | Sets $0.25t = \frac{5\pi}{2}$                                                                             | M1       | 3.1b         |
|           | 31.4                                                                                                      | A1       | 1.1b         |
|           |                                                                                                           | (2)      |              |
| (8 marks) |                                                                                                           |          |              |

(a)

**M1:** Separates the variables to reach  $\int \frac{1}{H} dH = \int \frac{\cos 0.25t}{40} dt$  or equivalent.

The integral signs need to be present on both sides and the  $dH$  AND  $dt$  need to be in the correct positions.

**M1:** Integrates both sides to reach  $\ln H = A \sin 0.25t$  or equivalent with or without the  $+c$

**A1:**  $\ln H = \frac{1}{10} \sin 0.25t + c$  or equivalent with or without the  $+c$ . Allow two constants, one either side

If the 40 was on the lhs look for  $40 \ln H = 4 \sin 0.25t + c$  or equivalent.

**dM1:** Substitutes  $t = 0, H = 5 \Rightarrow c = \dots$  There needs to have been a single  $+c$  to find.

It is dependent upon the previous M mark. You may allow even if you don't explicitly see  $t = 0, H = 5$  as it may be implied from their previous line

If the candidate has attempted to change the subject and made an error/ slip then condone it for this M but not the final A. Eg.  $40 \ln H = 4 \sin 0.25t + c \Rightarrow H^{40} = e^{4 \sin 0.25t} + e^c \Rightarrow 5^{40} = 1 + e^c \Rightarrow c = \dots$

Also many students will be attempting to get to the given answer so condone the method of finding  $c = \dots$   
These students will lose the A1\* mark

**A1\*:** Proceeds via  $\ln H = \frac{1}{10} \sin 0.25t + \ln 5$  or equivalent to the given answer  $H = 5e^{0.1 \sin 0.25t}$  with at least one correct intermediate line and no incorrect work.

DO NOT condone  $c$ 's going to  $c$ 's when they should be  $e^c$  or  $A$

Accept as a minimum  $\ln H = \frac{1}{10} \sin 0.25t + \ln 5 \Rightarrow H = e^{\frac{1}{10} \sin 0.25t + \ln 5}$  or  $H = e^{\frac{1}{10} \sin 0.25t} \times e^{\ln 5}$  before sight of the given answer

If the only error was to omit the integration signs on line 1, thus losing the first M1, allow the candidate to have access to this mark following a correct intermediate line (see above).

If they attempt to change the subject first then the constant of integration must have been adapted if the A1\* is to be awarded.  $\ln H = \frac{1}{10} \sin 0.25t + c \Rightarrow H = e^{\frac{1}{10} \sin 0.25t + c} \Rightarrow H = Ae^{\frac{1}{10} \sin 0.25t}$

The dM1 and A1\* under this method are awarded at virtually the same time.

Also, for the final two marks, you may see a proof from  $\int_0^H \frac{40}{H} dH = \int_5^t \cos 0.25t dt$

There is an alternative via the use of an integrating factor.

(b)

**B1:** States that the maximum height is 5.53 m Accept  $5e^{0.1}$  Condone a lack of units here, but penalise if incorrect units are used.

(c)

**M1:** For identifying that it would reach the maximum height for the 2nd time when  $0.25t = \frac{5\pi}{2}$  or 450

**A1:** Accept awrt 31.4 or  $10\pi$  Allow if units are seen

Q11.

| Part                   | Working or answer an examiner might expect to see                                                                         | Mark | Notes                                                                                                                        |
|------------------------|---------------------------------------------------------------------------------------------------------------------------|------|------------------------------------------------------------------------------------------------------------------------------|
| (a)                    | If $d = kV^n$ , then<br>$\log_{10} d = \log_{10} k + n \log_{10} V$                                                       | M1   | This mark is given for                                                                                                       |
|                        | Plotting $\log_{10} d$ against $\log_{10} V$ will result in a straight line with gradient $n$ and intercept $\log_{10} k$ | A1   | This mark is given for an explanation of why the second graph shows that $d = kV^n$                                          |
|                        | $\log_{10} k = -1.77$<br>$k = 10^{-1.77} = 0.01698 \dots \approx 0.017$                                                   | A1   | This mark is for showing fully that $k \approx 0.017$                                                                        |
| (b)                    | $d = kV^n$<br>When $V = 30$ , $d = 20$ and $k = 0.17$ then<br>$20 = 0.017 \times 30^n$                                    | M1   | This mark is given for substituting in the formula as a method to find the value of $n$                                      |
|                        | $n \log 30 = \log \left( \frac{20}{0.017} \right)$                                                                        | M1   | This mark is given for a correct expression for $n$                                                                          |
|                        | $n = 2.08$ to 3 significant figures<br>$d = 0.017 \times V^{2.08}$                                                        | A1   | This mark is given for finding a correct value of $n$ to 3 significant figures and writing a complete equation for the model |
| (c)                    | $\frac{60}{3600} \times 0.8 \times 1000 = 13.33 \text{ m}$                                                                | M1   | This mark is given for a method to find the distance, in metres, covered in the reaction time of 0.8 seconds                 |
|                        | $d = 0.017 \times 60^{2.08} = 84.92 \text{ m}$                                                                            | M1   | This mark is given for a method to use the formula to find the stopping distance                                             |
|                        | $13.33 \text{ m} + 84.92 \text{ m} = 98.25 \text{ m}$<br>Sean will be able to stop before reaching the puddle             | A1   | This mark is given for finding a correct value of the total stopping distance and giving a valid conclusion                  |
| <b>(Total 9 marks)</b> |                                                                                                                           |      |                                                                                                                              |



Q12.

| Question  | Scheme                                                                                                                                                                                                                                                                                               | Marks    | AOs          |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------|
| (a)       | $f(x) = 4(x^2 - 2)e^{-2x}$                                                                                                                                                                                                                                                                           |          |              |
|           | Differentiates to $e^{-2x} \times 8x + 4(x^2 - 2) \times -2e^{-2x}$                                                                                                                                                                                                                                  | M1<br>A1 | 1.1b<br>1.1b |
|           | $f'(x) = 8e^{-2x} \{x - (x^2 - 2)\} = 8(2 + x - x^2)e^{-2x}$ *                                                                                                                                                                                                                                       | A1*      | 2.1          |
|           |                                                                                                                                                                                                                                                                                                      | (3)      |              |
| (b)       | States roots of $f'(x) = 0$ $x = -1, 2$                                                                                                                                                                                                                                                              | B1       | 1.1b         |
|           | Substitutes one $x$ value to find a $y$ value                                                                                                                                                                                                                                                        | M1       | 1.1b         |
|           | Stationary points are $(-1, -4e^2)$ and $(2, 8e^{-4})$                                                                                                                                                                                                                                               | A1       | 1.1b         |
|           |                                                                                                                                                                                                                                                                                                      | (3)      |              |
| (c)       | (i) Range $[-8e^2, \infty)$ o.e. such as $g(x) \geq -8e^2$                                                                                                                                                                                                                                           | B1ft     | 2.5          |
|           | (ii) For <ul style="list-style-type: none"> <li>Either attempting to find <math>2f(0) - 3 = 2 \times -8 - 3 = (-19)</math> and identifying this as the <b>lower bound</b></li> <li>Or attempting to find <math>2 \times 8e^{-4} - 3</math> and identifying this as the <b>upper bound</b></li> </ul> | M1       | 3.1a         |
|           | Range $[-19, 16e^{-4} - 3]$                                                                                                                                                                                                                                                                          | A1       | 1.1b         |
|           |                                                                                                                                                                                                                                                                                                      | (3)      |              |
| (9 marks) |                                                                                                                                                                                                                                                                                                      |          |              |
| Notes:    |                                                                                                                                                                                                                                                                                                      |          |              |

- (a)
- M1:** Attempts the product rule and uses  $e^{-2x} \rightarrow ke^{-2x}$ ,  $k \neq 0$
- If candidate states  $u = 4(x^2 - 2)$ ,  $v = e^{-2x}$  with  $u' = \dots$ ,  $v' = \dots e^{-2x}$  it can be implied by their  $vu' + uv'$
- If they just write down an answer without working award for  $f'(x) = pxe^{-2x} \pm q(x^2 - 2)e^{-2x}$
- They may multiply out first  $f(x) = 4x^2e^{-2x} - 8e^{-2x}$ . Apply in the same way condoning slips
- Alternatively attempts the quotient rule on  $f(x) = \frac{u}{v} = \frac{4(x^2 - 2)}{e^{2x}}$  with  $v' = ke^{2x}$  and  $f'(x) = \frac{vu' - uv'}{v^2}$
- A1:** A correct  $f'(x)$  which may be unsimplified.
- Via the quotient rule you can award for  $f'(x) = \frac{8xe^{2x} - 8(x^2 - 2)e^{2x}}{e^{4x}}$  o.e.
- A1\*:** Proceeds correctly to given answer showing all necessary steps.
- The  $f'(x)$  or  $\frac{dy}{dx}$  must be present at some point in the solution
- This is a "show that" question and there must not be any errors. All bracketing must be correct.
- Allow a candidate to move from the **simplified** unfactorised answer of  $f'(x) = 8xe^{-2x} - 8(x^2 - 2)e^{-2x}$



to the given answer in one step.

Do not allow it from an **unsimplified**  $f'(x) = 4 \times 2xe^{-2x} + 4(x^2 - 2) \times -2e^{-2x}$

Allow the expression / bracketed expression to be written in a different order.

So, for example,  $8(x - x^2 + 2)e^{-2x}$  is OK

(b)

**B1:** States or implies  $x = -1, 2$  (as the roots of  $f'(x) = 0$ )

**M1:** Substitutes one  $x$  value of their solution to  $f'(x) = 0$  in  $f(x)$  to find a  $y$  value.

Allow decimals here (3sf). FYI, to 3 sf,  $-4e^2 = -29.6$  and  $8e^{-4} = 0.147$

Some candidates just write down the  $x$  coordinates but then go on in part (c) to find the ranges using the  $y$  coordinates. Allow this mark to be scored from work in part (c)

**A1:** Obtains  $(-1, -4e^2)$  and  $(2, 8e^{-4})$  as the stationary points. This must be scored in (b). Remember to isw

after a correct answer. Allow these to be written separately. E.g.  $x = -1, y = -4e^2$

Extra solutions, e.g. from  $x = 0$  will be penalised on this mark.

(c)(i)

**B1ft:** For a correct range written using correct notation.

Follow through on  $2 \times$  their minimum "y" value from part (b), providing it is negative.

Condone a decimal answer if this is consistent with their answer in (b) to 3sf or better.

Examples of correct responses are  $[-8e^2, \infty)$ ,  $g \geq -8e^2$ ,  $y \geq -8e^2$ ,  $\{q \in \mathbb{R}, q \geq -8e^2\}$

(c)(ii)

**M1:** See main scheme. Follow through on  $2 \times$  their " $8e^{-4}$ "  $-3$  for the upper bound.

**A1:** Range  $[-19, 16e^{-4} - 3]$  o.e. such as  $-19 \leq y \leq 16e^{-4} - 3$  but must be exact

Q13.

| Question | Scheme                                                                                        | Marks | AOs  |
|----------|-----------------------------------------------------------------------------------------------|-------|------|
| (a)      | $N = aT^b \Rightarrow \log_{10} N = \log_{10} a + \log_{10} T^b$                              | M1    | 2.1  |
|          | $\Rightarrow \log_{10} N = \log_{10} a + b \log_{10} T$ so $m = b$ and $c = \log_{10} a$      | A1    | 1.1b |
|          |                                                                                               | (2)   |      |
| (b)      | Uses the graph to find either $a$ or $b$ $a = 10^{\text{intercept}}$ or $b = \text{gradient}$ | M1    | 3.1b |
|          | Uses the graph to find both $a$ and $b$ $a = 10^{\text{intercept}}$ and $b = \text{gradient}$ | M1    | 1.1b |
|          | Uses $T = 3$ in $N = aT^b$ with their $a$ and $b$                                             | M1    | 3.1b |
|          | Number of microbes $\approx 800$                                                              | A1    | 1.1b |
|          |                                                                                               | (4)   |      |
| (c)      | $N = 1000000 \Rightarrow \log_{10} N = 6$                                                     | M1    | 3.4  |
|          | We cannot 'extrapolate' the graph and assume that the model still holds                       | A1    | 3.5b |
|          |                                                                                               | (2)   |      |
| (d)      | States that ' $a$ ' is the number of microbes 1 day after the start of the experiment         | B1    | 3.2a |
|          |                                                                                               | (1)   |      |

(9 marks)

**Notes:**

(a)

M1: Takes logs of both sides and shows the addition law

M1: Uses the power law, writes  $\log_{10} N = \log_{10} a + b \log_{10} T$  and states  $m = b$  and  $c = \log_{10} a$

(b)

M1: Uses the graph to find either  $a$  or  $b$   $a = 10^{\text{intercept}}$  or  $b = \text{gradient}$ . This would be implied by the sight of  $b = 2.3$  or  $a = 10^{1.8} \approx 63$

M1: Uses the graph to find both  $a$  and  $b$   $a = 10^{\text{intercept}}$  and  $b = \text{gradient}$ . This would be implied by the sight of  $b = 2.3$  and  $a = 10^{1.8} \approx 63$

M1: Uses  $T = 3 \Rightarrow N = aT^b$  with their  $a$  and  $b$ . This is implied by an attempt at  $63 \times 3^{2.3}$

A1: Accept a number of microbes that are approximately 800. Allow  $800 \pm 150$  following correct work.

There is an alternative to this using a graphical approach.

M1: Finds the value of  $\log_{10} T$  from  $T = 3$ . Accept as  $T = 3 \Rightarrow \log_{10} T \approx 0.48$

M1: Then using the line of best fit finds the value of  $\log_{10} N$  from their "0.48"  
Accept  $\log_{10} N \approx 2.9$

M1: Finds the value of  $N$  from their value of  $\log_{10} N$   $\log_{10} N \approx 2.9 \Rightarrow N = 10^{2.9}$

A1: Accept a number of microbes that are approximately 800. Allow  $800 \pm 150$  following correct work

(c)

**M1** For using  $N = 1000000$  and stating that  $\log_{10} N = 6$

**A1:** Statement to the effect that "we only have information for values of  $\log N$  between 1.8 and 4.5 so we cannot be certain that the relationship still holds". "We cannot extrapolate with any certainty, we could only interpolate"

There is an alternative approach that uses the formula.

**M1:** Use  $N = 1000000$  in their  $N = 63 \times T^{2.3} \Rightarrow \log_{10} T = \frac{\log_{10} \left( \frac{1000000}{63} \right)}{2.3} \approx 1.83$ .

**A1:** The reason would be similar to the main scheme as we only have  $\log_{10} T$  values from 0 to 1.2. We cannot 'extrapolate' the graph and assume that the model still holds

(d)

**B1:** Allow a numerical explanation  $T = 1 \Rightarrow N = a1^b \Rightarrow N = a$  giving  $a$  is the value of  $N$  at  $T = 1$